MATH329 · OPERATIONS RESEARCH

Midterm 461 Solutions

10/10/2024 · 25 marks

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01 Solution Q1: theory

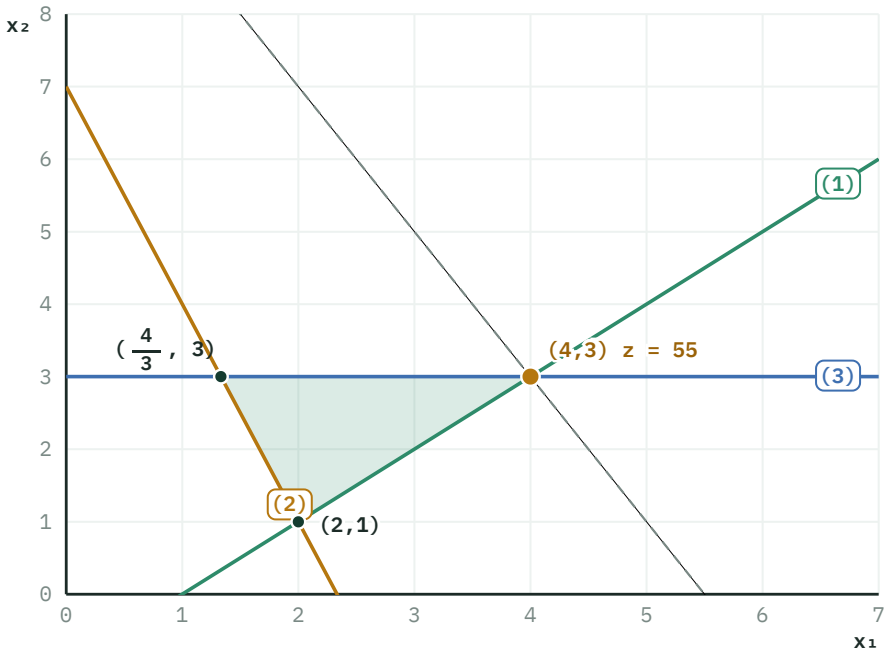
#	Answer
a	**England**
b	**Linear Programming Problem (LPP)**
c	1. All decision variables are **nonnegative**. 2. All constraints (with the exception of the non-negativity of the variables) are **equations with nonnegative RHS**.

02 Solution Q2: graphical

1 Prepare the constraints

(1)	$x_1 - x_2 = 1$	points (1, 0) and (0, −1)
(2)	$3x_1 + x_2 = 7$	points (0, 7) and $(\frac{7}{3}, 0)$
(3)	$x_2 = 3$	horizontal line

Testing `(0, 0)`: constraint (1) is true, so its region is on the origin's side. Constraint (2) is false because `0 ≥ 7` does not hold, so its region is on the far side. Constraint (3) is below its line.



2 The corner points

Corner	Where it comes from	$z = 10x_1 + 5x_2$
(2, 1)	(1) with (2)	25
(4, 3)	(1) with (3)	**55**
$(\frac{4}{3}, 3)$	(2) with (3)	28.33

(1) with (3): $x_1 - 3 = 1 \rightarrow x_1 = 4$

$Z = 55$ at $x_1 = 4, x_2 = 3$

The region here is a small triangle that does not touch the axes, because constraint (2) pushes it away from the origin.

03 Solution Q3: standard form

1 The methods 1 M

Graphical method and Algebraic method.

2 Standard form

$4x_2 = -3 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 3$

$x_2 = x_2^+ - x_2^- \quad$ in every place x_2 appears

$$
\begin{aligned}
&\text{Minimize } Z = 7x_1 + 3x_2^+ - 3x_2^- + 0s_1 + 0s_2 \\
&\text{S.to} \quad 2x_1 + x_2^+ - x_2^- + s_1 = 8 \\
&\qquad x_1 - 3x_2^+ + 3x_2^- - s_2 = 12 \\
&\qquad -4x_2^+ + 4x_2^- = 3 \\
&\qquad x_1, x_2^+, x_2^-, s_1, s_2 \geq 0
\end{aligned}
$$

An error in the answer key

The key writes the second constraint's right-hand side as `14`, while the question has `12`. The conversion does not change the right-hand side, so `12` is correct.

04 Solution Q4: standard form and simplex

a Standard form 2.5 M

$$
\begin{aligned}
&\text{Maximize } Z = 8x_1 + 7x_2 + 0s_1 + 0s_2 + 0s_3 \\
&\text{S.to} \quad 20x_1 + 16x_2 + s_1 = 62 \\
&\qquad -x_1 + x_2 + s_2 = 1 \\
&\qquad x_1 + s_3 = 3 \\
&\qquad x_1, x_2, s_1, s_2, s_3 \geq 0
\end{aligned}
$$

b **Simplex** 2 TABLES ONLY

Basic	x_1	x_2	s_1	s_2	s_3	Sol.	Ratio
Z	−8	−7	0	0	0	0	
s_1	20	16	1	0	0	62	$\frac{31}{10}$
s_2	−1	1	0	1	0	1	–
s_3	**1**	0	0	0	1	3	3
Z	0	−7	0	0	8	24	
s_1	0	16	1	0	−20	2	
s_2	0	1	0	1	1	4	
x_1	1	0	0	0	1	3	

The most negative entry is -8, so x_1 enters. The ratios are $\frac{62}{20} = 3.1$, none for the second row (negative coefficient), and $\frac{3}{1} = 3$. The smallest is 3, so s_3 leaves and the pivot is 1.

New x_1 row = s_3 row ÷ 1 = (1, 0, 0, 0, 1 | 3)

New Z row = Z row − (−8) × new x_1 row = (0, −7, 0, 0, 8 | 24)

New s_1 row = s_1 row − 20 × new x_1 row = (0, 16, 1, 0, −20 | 2)

New s_2 row = s_2 row − (−1) × new x_1 row = (0, 1, 0, 1, 1 | 4)

We stop here because the question asks for two tables only.

First table 2m (every row 0.5m + 0.5m for pivot element).
Second table 4m (1m for every row). Every mistake in calculation −0.5m.

Notice the pattern

The 472 question is a copy of this one with numbers roughly doubled: the same three constraints, the same entering and leaving variables, and the same pivot position.